

ZHIHUI CHEN

Phone: (+86) 13310890335 | Email: zhihui.chen@u.nus.edu
Target Role: LLM Research Intern | Homepage: richardchenzhihui.github.io



Education

National University of Singapore Ph.D. Student in Artificial Intelligence • Research Focus: Multimodal large language models, medical AI, and trustworthy and responsible AI.	2025.01 - 2028.12
The University of Hong Kong M.Sc. in Artificial Intelligence	2022.09 - 2024.07
The Chinese University of Hong Kong, Shenzhen B.Sc. in Statistics, Data Science Track	2018.09 - 2022.05

Research Interests

Post-training for multimodal large language models • Medical AI and medical foundation models • Trustworthiness / alignment • Reinforcement learning for preference modeling

Publications (First Author)

- [Zhihui Chen](#), Kai He, Qingyuan Lei, Mengling Feng. "MedForge: Interpretable Medical Deepfake Detection via Forgery-aware Reasoning". **ACL 2026 Main Conference**
- [Zhihui Chen](#), Kai He, Yucheng Huang, Yunxiao Zhu, Mengling Feng. "DivScore: Zero-Shot Detection of LLM-Generated Text in Specialized Domains." **EMNLP 2025 Main Conference**
- [Zhihui Chen](#), Mengling Feng, et al. "Med-Banana: Agentic Quality-Controlled Medical Image Editing from Success-and-Failure Trajectories". EMNLP 2026 submission [OA3.17 Top 10%](#)
- [Zhihui Chen](#), Yike Yun et al. "Looking Ahead to Stay Grounded: Future-Discounted Visual Credit Assignment for VLM Reinforcement Learning". ICLR 2026 submission

Academic Service

Reviewer for ACL 2026, NeurIPS 2026, IJCAI 2025, IEEE Affective Computing, ACM TIST, and other top conferences and journals, with 40+ reviews completed.

Research Experience

MedForge: Interpretable Medical Deepfake Detection

- Built the MedForge-90K benchmark dataset, covering 19 disease categories across CXR, MRI, fundus imaging, and more, with 90,000 images and hierarchical expert chain-of-thought annotations.
- Built a CoT medical deepfake expert dataset that identifies tampering in medical images through a "localize first, then reason" mechanism, providing interpretable reasoning at the chain-of-thought level.
- Proposed Forgery-aware GSPO to constrain model reasoning with visual bounding-box evidence, reducing hallucinations by 16.2% and producing explanations with stronger clinically verifiable professionalism.

Med-Banana-50K: Multi-Agent System for Medical Image Editing

- Built the first large-scale instruction-based medical image editing dataset, Med-Banana-50K, covering 50,000 high-quality editing pairs across chest X-rays, brain MRI, fundus images, and more, and organized "accepted/rejected candidates" into chosen-rejected preference pairs.
- Designed an editor-verifier-refiner multi-agent quality-control pipeline, combining LLM-as-Judge with medical-specific rubrics for perceptual iterative validation around rubric-guided criteria.
- Based on 50,000 success trajectories and 37,000 failure negative samples, conducted two-stage SFT + Diffusion-DPO training on an open-source diffusion editor, improving editing fidelity and robustness in multimodal generation.

DivScore: Zero-Shot LLM Text Detection

- Proposed the DivScore zero-shot text detection framework, using normalized entropy scoring and unsupervised domain knowledge distillation to identify LLM-generated text in specialized domains.
- Achieved SOTA on medical and legal benchmarks, improving AUROC by 14.4% over the best baseline and recall by 64.0% at 0.1% FPR.
- Derived the distributional differences between human-written and LLM-generated text from an information-theoretic perspective, demonstrating the effectiveness of entropy normalization under domain transfer.

Internships and Project Experience

ByteDance R&D Intern (Multimodal Agent Reinforcement Learning)2026.06 - 2026.11

R&D Intern, ByteDance Singapore

- Proposed V-FIPO for RLVR post-training of VLMs, extending token-level credit assignment from "instant visual dependence" to "discounted future visual grounding", with a first-author ICLR 2026 submission.
- Contributed to a multimodal e-commerce livestream moderation agent, supporting training-data construction, RLVR reward design, and evaluation-loop setup around image/video understanding, tool use, and long-horizon task completion.
- Managed trajectory data cleaning, preference-data organization, and sample-quality analysis, and helped build task replay, result verification, and failure-mode attribution workflows to support high-quality data for reinforcement learning iterations.

StepFun R&D Intern (StepAudio2.5 ASR Speech Foundation Model)2025.03 - 2025.07

R&D Intern, Shanghai StepFun Intelligent Technology Co., Ltd.

- Managed data governance including audio segmentation, transcription alignment, duplicate removal, low-quality filtering, dialect cleaning and classification for Hakka, Teochew, and Cantonese, as well as normalization of emotion and scenario labels, improving the consistency, controllability, and reusability of pretraining corpora. The work contributed to StepAudio2.5 ASR.
- Designed and maintained pretraining-data organization schemes and metadata to support retrieval, sampling, quality inspection, and version iteration across dialect and emotional-style control dimensions, providing a stable data foundation for later model training, evaluation, and traceability.
- Participated in the construction of the speech training-data system for ASR, TTS, and real-time speech interaction, integrating open-source and internal corpora and completing speech-text paired data ingestion.

Quant Trading Agent: Hong Kong Stock Quantitative Agent2025.01 - 2025.03

Quantitative Trading R&D Intern, AQUOMON (Hong Kong)

- Participated in the development of a quantitative trading agent based on the LangChain architecture, connecting core workflows including market understanding, signal generation, strategy decision-making, and execution monitoring to support automated operation in Hong Kong stock programmatic trading scenarios.
- Integrated Futu OpenAPI for real-time market data access and simulated automated order execution, and helped build strategy validation and integration-testing workflows to improve the efficiency of strategy iteration and deployment.

Office Docx Harness Agent: Intelligent Document Editing Assistant2025.01 - 2025.02

Author / Lead Developer, GitHub Open-Source Project

- Independently designed and implemented a model-agnostic agent harness centered on a ReAct loop, packaging Word document parsing, editing, and reflow into a schedulable tool layer, compiling natural-language instructions into action sequences with execution, verification, and persistence, covering composite edits such as bulk replacement, formatting standardization, paragraph/table insertion and deletion, and image processing.
- Built a multi-document context fusion mechanism that uses attachments as reference knowledge to drive batch editing of the main document; the model is decoupled from the harness and hot-swappable, with backend compatibility for OpenAI / OpenRouter.

Teaching Experience

Undergraduate Course Teaching Assistant: "Generative AI and LLM"2024.02 - 2024.04

NUS Business School, Prof. Pang Yan

- Participated in designing the course knowledge framework and teaching content.
- Wrote lecture notes and course projects covering PyTorch, HuggingFace, LLM pretraining, SFT, RLHF, and related topics.

Awards

- Full Ph.D. Scholarship, National University of Singapore (2025 - 2029)
- Outstanding Graduate, The Chinese University of Hong Kong, Shenzhen (Top 5%)
- Undergraduate Research Excellence Award, The Chinese University of Hong Kong, Shenzhen

Skills

Programming	Python, R, C++, MATLAB, MySQL
Deep Learning Frameworks	PyTorch, TensorFlow, scikit-learn, HuggingFace
AI Tools	Linux, Docker, Git, Llama.cpp
Languages	English (TOEFL 106, GRE 320), Mandarin (Native), Cantonese (Native)

Interests: Photography, Cycling, Violin, Running